# Masked observations

## 1 Masked observations

**Theorem 1.1.** *For every natural number $t$,* $\mathrm{whitecount}(t, 0) = 0$.

*Proof.* The goal follows from $\mathrm{whitecountzero}(t)$. □

**Theorem 1.2.** *For every natural number $t$ and natural number $n$, if* $\mathrm{center}(t) = \mathrm{true}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n)$.

*Proof.* The goal follows from $\mathrm{whitecountblackstep}(t, n)$. □

**Theorem 1.3.** *For every natural number $t$ and natural number $n$, if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n) + 1$.

*Proof.* The goal follows from $\mathrm{whitecountwhitestep}(t, n)$. □

**Theorem 1.4.** *For every natural number $t1$ and natural number $t2$ and natural number $n$, if* $\mathrm{centerwindowagree}(t1, t2, n)$*, then* $\mathrm{leftprefixagree}(t1, t2, n)$ *if and only if* $\mathrm{whitetraceagree}(t1, t2, n)$.

*Proof.* The goal follows from $\mathrm{prefixagreementiffwhitetrace}(t1, t2, n)$. □

**Theorem 1.5.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then there exists a natural number $a$ such that there exists a natural number $b$ such that $a < b$ and $b \leq$* $\mathrm{twopow}(\mathrm{whitecount}(t, n)) \wedge \mathrm{leftprefixagree}(t + a \cdot P, t + b \cdot P, n)$.

*Proof.* The goal follows from $\mathrm{compressedprefixrepeat}(T, P, t, n)$. □

**Theorem 1.6.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$ and natural number $M$, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then if* $\mathrm{prefixseparated}(t, P, n, M)$*, then* $M \leq \mathrm{twopow}(\mathrm{whitecount}(t, n))$.

*Proof.* The goal follows from $\mathrm{separatedprefixbound}(T, P, t, n, M)$. □

**Theorem 1.7.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then* $n < t + \mathrm{twopow}(\mathrm{whitecount}(t, n)) \cdot P$.

*Proof.* The goal follows from $\mathrm{edgeinformationbound}(T, P, t, n)$. □

**Theorem 1.8.** *For every natural number $n$ and natural number $t$ and natural number $w$ and natural number $P$, if $n < (w+1) \cdot P$, then $n < t + \mathrm{twopow}(w) \cdot P$.*

*Proof.* The goal follows from $\mathrm{informationboundaudit}(n, t, w, P)$. □

## 2 Central collision recurrence

**Theorem 2.1.** *For every natural number $Q$ and natural number $t$ and natural number $r$, if $\mathrm{leftadjacentdefect}(Q, t) = \mathrm{true}$, then if $\mathrm{center}(t + r) = \mathrm{true}$, then if $\mathrm{center}(t + r + Q) = \mathrm{true}$, then $\mathrm{nearcollision}(Q, t, r + 1, 2 \cdot (r + 1))$.*

*Proof.* The goal follows from $\mathrm{unitdefectforcesnearcollision}(Q, t, r)$. □

**Theorem 2.2.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq t$, then if $\mathrm{leftadjacentdefect}(m \cdot P, t) = \mathrm{true}$, then $\mathrm{nearcollision}(m \cdot P, t, P, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodicunitdefectforcesnearcollision}(T, P, m, t)$. □

**Theorem 2.3.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + P \wedge \mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodiccentralcollisionwindow}(T, P, m, N)$. □

**Theorem 2.4.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then there exists a natural number $t$ such that $N \leq t$ and $\mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodiccentralcollisions}(T, P, m, N)$. □

**Theorem 2.5.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then $\mathrm{recurrentcentralcollision}(m \cdot P, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodicrecurrentcentralcolumn}(T, P, m)$. □

**Theorem 2.6.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*leftcollision$(Q, b+s, r+1)$.

*Proof.* The goal follows from blackwindowforcesleftcollision$(Q, b, r)$. □

**Theorem 2.7.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* leftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicleftcollisionwindow$(T, P, m, N)$. □

**Theorem 2.8.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then there exists a natural number $t$ such that $N \leq t$ and* leftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicleftcollisions$(T, P, m, N)$. □

**Theorem 2.9.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentleftcollision$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentleftcolumn$(T, P, m)$. □

**Theorem 2.10.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* collisioncode$(Q, t, k) \neq 0$ *if and only if* collisionbit$(Q, t, k) =$ true.

*Proof.* The goal follows from collisioncodenonzero$(Q, t, k)$. □

**Theorem 2.11.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* collisioncode$(Q, t, k) \leq 6$.

*Proof.* The goal follows from collisioncodebound$(Q, t, k)$. □

**Theorem 2.12.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* collisioncode$(Q, t, k) = 1 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 2$ *and* (collisioncode$(Q, t, k) = 2 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 3 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 4 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 5 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 6 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$ leftcolour$(Q, t, k - 1) = 2)$.

*Proof.* The goal follows from collisioncodecolours$(Q, t, k)$. □

**Theorem 2.13.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* advancingatdepth$(Q, t, k)$ *if and only if* collisioncode$(Q, t, k) = 3$ *or* collisioncode$(Q, t, k) = 5$.

*Proof.* The goal follows from advancingcollisioncodes$(Q, t, k)$. □

**Theorem 2.14.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftdefect$(Q, t + 1, k) =$ leftdefect$(Q, t, k + 1)$.

*Proof.* The goal follows from collisiontransmitsleft$(Q, t, k)$. □

**Theorem 2.15.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftgatedifference$(Q, t, k) =$ false.

*Proof.* The goal follows from collisiongatesagree$(Q, t, k)$. □

**Theorem 2.16.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $2 \leq k$, then* collisioncode$(Q, t, k) = 1 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t, k-2)$ *and* (collisioncode$(Q, t, k) = 2 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t+Q, k-2)) \wedge$(collisioncode$(Q, t, k) = 3 \to$ leftdefect$(Q, t+1, k-1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 4 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t + Q, k - 2))) \wedge$ (collisioncode$(Q, t, k) = 5 \to$ leftdefect$(Q, t + 1, k - 1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 6 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t, k - 2))$).

*Proof.* The goal follows from collisionrighttransporttable$(Q, t, k)$. □

**Theorem 2.17.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* collisioncode$(Q, t, k) = 3 \to$ leftcolour$(Q, t + 1, k-1) = 2$ *and if* collisioncode$(Q, t, k) = 5$*, then* leftcolour$(Q, t+1, k-1) =$ 1.

*Proof.* The goal follows from advancingrightcolourtable$(Q, t, k)$. □

**Theorem 2.18.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftdefect$(Q, t + 1, k) =$ not(leftdefect$(Q, t, k + 1)$) *if and only if* leftgatedifference$(Q, t, k) =$ true.

*Proof.* The goal follows from gatedifferenceiffflip$(Q, t, k)$. □

**Theorem 2.19.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k+1) =$ true*, then* leftdefect$(Q, t+1, k) =$ false *if and only if* leftgatedifference$(Q, t, k) =$ true*.*

*Proof.* The goal follows from incomingdefecterasediff$(Q, t, k)$. □

**Theorem 2.20.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k+1) =$ true*, then if* leftdefect$(Q, t+1, k) =$ false*, then* collisioncode$(Q, t, k) = 0$*.*

*Proof.* The goal follows from erasurehasnocollision$(Q, t, k)$. □

**Theorem 2.21.** *For every natural number $Q$ and natural number $k$, if $0 < k$, then* leftcolour$(Q, 0, k) = 0$ *or* leftcolour$(Q, 0, k) = 2$*.*

*Proof.* The goal follows from seedleftcolours$(Q, k)$. □

**Theorem 2.22.** *For every natural number $Q$ and natural number $k$, if $2 \leq k$, then* collisioncode$(Q, 0, k) = 0$*.*

*Proof.* The goal follows from seedleftcollisionzero$(Q, k)$. □

**Theorem 2.23.** collisioncode$(1, 1, 2) = 5$*.*

*Proof.* The goal follows from seedadvancingcollisionbirth. □

**Theorem 2.24.** leftdefect$(3, 0, 3) =$ true *and* leftdefect$(3, 1, 2) =$ false $\wedge$ leftgatedifference$(3, 0, 2) =$ true*.*

*Proof.* The goal follows from seedgateerasure. □

**Theorem 2.25.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k) =$ true*, then if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then* advancingleftcone$(Q, t, k)$*.*

*Proof.* The goal follows from leftdefectblackcone$(Q, t, k)$. □

**Theorem 2.26.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*advancingleftcollision$(Q, b+s, r+1)$*.*

*Proof.* The goal follows from blackwindowforcesadvancingcollision$(Q, b, r)$. □

**Theorem 2.27.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* advancingleftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicadvancingleftcollisionwindow$(T, P, m, N)$. □

**Theorem 2.28.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentadvancingleftcollision$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentadvancingleftcolumn$(T, P, m)$. □

**Theorem 2.29.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftdefect$(Q, t, k) =$ false *if and only if* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 3$.

*Proof.* The goal follows from leftdefectfalsecolours$(Q, t, k)$. □

**Theorem 2.30.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftdefect$(Q, t, k) =$ true *if and only if* leftcolour$(Q, t, k) = 1$ *or* leftcolour$(Q, t, k) = 2$.

*Proof.* The goal follows from leftdefecttruecolours$(Q, t, k)$. □

**Theorem 2.31.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $t < k$, then* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 2$.

*Proof.* The goal follows from leftcoloursoutsideseedcone$(Q, t, k)$. □

**Theorem 2.32.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then $k \leq t + 1$.*

*Proof.* The goal follows from collisioncodesupport$(Q, t, k)$. □

**Theorem 2.33.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* gatecode$(Q, t, k) \neq 0$ *if and only if* leftgatedifference$(Q, t, k) =$ true.

*Proof.* The goal follows from gatecodenonzero$(Q, t, k)$. □

**Theorem 2.34.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{gatecode}(Q, t, k) \leq 6$.

*Proof.* The goal follows from $\mathrm{gatecodebound}(Q, t, k)$. □

**Theorem 2.35.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* $\mathrm{gatecode}(Q, t, k) = 1 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0$ *and* $(\mathrm{gatecode}(Q, t, k) = 2 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 3 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 4 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0) \wedge (\mathrm{gatecode}(Q, t, k) = 5 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2) \wedge (\mathrm{gatecode}(Q, t, k) = 6 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2)$.

*Proof.* The goal follows from $\mathrm{gatecodecolours}(Q, t, k)$. □

**Theorem 2.36.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftdefect}(Q, t, k+1) = \mathrm{true}$*, then if* $\mathrm{leftdefect}(Q, t+1, k) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, k)$.

*Proof.* The goal follows from $\mathrm{erasureeventofstep}(Q, t, k)$. □

**Theorem 2.37.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 3$ *if and only if* $\mathrm{gatecode}(Q, t, k) \leq 3$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 2$ *or* $4 \leq \mathrm{gatecode}(Q, t, k)$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 1$.

*Proof.* The goal follows from $\mathrm{erasureblackphaseiff}(Q, t, k)$. □

**Theorem 2.38.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $t+1 < k$, then if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 0$.

*Proof.* The goal follows from $\mathrm{erasureoutsideseedconewhite}(Q, t, k)$. □

**Theorem 2.39.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $2 \leq k$, then if* $\mathrm{advancingatdepth}(Q, t, k)$*, then if* $\mathrm{leftdefect}(Q, t+k, 0) = \mathrm{false}$*, then there exists a natural number $j$ such that* $\mathrm{firstpatherasure}(Q, t, k, j)$.

*Proof.* The goal follows from $\mathrm{advancingsourcefirsterasure}(Q, t, k)$. □

**Theorem 2.40.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then there exists a natural number $u$ such that there exists a natural number $d$ such that $N \leq u$ and $u + 1 \leq 2 \cdot N + m \cdot P + 3 \cdot P \wedge d + 2 \leq P \wedge$* $\mathrm{erasureevent}(m \cdot P, u, d)$.

*Proof.* The goal follows from periodicerasurewindow$(T, P, m, N)$. □

**Theorem 2.41.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$, *then* recurrentsourceerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentsourceerasure$(T, P, m)$. □

**Theorem 2.42.** noperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from noperiodiffnorecurrentcentralcollision. □

**Theorem 2.43.** norowcenterperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from norowcenterperiodiffnorecurrentcentralcollision. □

**Theorem 2.44.** problemone *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from problemoneiffnorecurrentcentralcollision. □

**Theorem 2.45.** *If* norecurrentcentralcollision, *then* noperiod.

*Proof.* Assume $h$. The goal follows from noperiodofnorecurrentcentralcollision$(h)$. □

**Theorem 2.46.** *For every natural number $T$ and natural number $P$, if $0 < P$, then if* norecurrentcentralcollision, *then not* centerperiodic$(T, P)$.

*Proof.* The goal follows from notcenterperiodicofnorecurrentcentralcollision$(T, P)$. □

**Theorem 2.47.** *If* norecurrentcentralcollision, *then* problemone.

*Proof.* Assume $h$. The goal follows from noteventuallyperiodicofnorecurrentcentralcollision$(h)$. □

**Theorem 2.48.** *For every natural number $b$ and natural number $d$ and natural number $i$,* pairphase(wittphi$(b)$, wittxor(wittphi$(d)$, wittshl(crosscarry$(b, d), 1)), i) =$ bitvalue(rowbit(wittphi(wittxor$(b, d)), i)) -$ bitvalue(rowbit(wittphi$(b), i)$).

*Proof.* The goal follows from pairphasestep$(b, d, i)$. □

**Theorem 2.49.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* alignedphase$(Q, t, t + k + Q) =$ leftphase$(Q, t, k)$.

*Proof.* The goal follows from alignedphaseleft$(Q, t, k)$. □

**Theorem 2.50.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k) = 0$ *if and only if* leftdefect$(Q, t, k) =$ false*.*

*Proof.* The goal follows from phasezeroiff$(Q, t, k)$. □

**Theorem 2.51.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k) = -1$ *or* leftphase$(Q, t, k) = 1$ *if and only if* leftdefect$(Q, t, k) =$ true*.*

*Proof.* The goal follows from phaseunitiff$(Q, t, k)$. □

**Theorem 2.52.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t+1, k) =$ leftphase$(Q, t, k+1)+$leftphaseflux$(Q, t, k)$.

*Proof.* The goal follows from leftphasebalance$(Q, t, k)$. □

**Theorem 2.53.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftphaseflux$(Q, t, k) = -2 \cdot$ leftphase$(Q, t, k + 1)$*.*

*Proof.* The goal follows from collisionphaseflip$(Q, t, k)$. □

**Theorem 2.54.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then if* advancingatdepth$(Q, t, k)$*, then* leftphase$(Q, t+1, k-1) = -$ leftphase$(Q, t, k)$*.*

*Proof.* The goal follows from advancingphaseflip$(Q, t, k)$. □

**Theorem 2.55.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* gatecode$(Q, t, k) = 1 \to$ leftgatephase$(Q, t, k) = -1$ *and* (gatecode$(Q, t, k) = 2 \to$ leftgatephase$(Q, t, k) = -1) \wedge$ (gatecode$(Q, t, k) = 3 \to$ leftgatephase$(Q, t, k) = -1) \wedge$(gatecode$(Q, t, k) = 4 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 5 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 6 \to$ leftgatephase$(Q, t, k) = 1)$.

*Proof.* The goal follows from gatephasecases$(Q, t, k)$. □

**Theorem 2.56.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$*, then* leftphase$(Q, t + 1, k) = 0$ *and* leftphaseflux$(Q, t, k) = -$ leftphase$(Q, t, k+1) \wedge$ (leftphase$(Q, t, k+1) = -1 \vee$ leftphase$(Q, t, k + 1) = 1)$*.*

*Proof.* The goal follows from erasurephasebalance$(Q, t, k)$. □

**Theorem 2.57.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$*, then* leftcolour$(Q, t+1, k) = 3 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$leftphase$(Q, t, k+1) = -1$ *and* leftcolour$(Q, t+1, k) = 0 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$ leftphase$(Q, t, k+1) = 1$.

*Proof.* The goal follows from erasureresetproduct$(Q, t, k)$. □

**Theorem 2.58.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $n$,* pathphase$(Q, t, k, n) =$ leftphase$(Q, t, k)+$ pathflux$(Q, t, k, n)$.

*Proof.* The goal follows from pathphasebalance$(Q, t, k, n)$. □

**Theorem 2.59.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* leftphase$(Q, t, t + Q) = 1$.

*Proof.* The goal follows from seedleftphase$(Q, t)$. □

**Theorem 2.60.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* leftphase$(Q, t + (t + Q), 0) = 1 +$ pathflux$(Q, t, t + Q, t + Q)$.

*Proof.* The goal follows from seedcenterphasebalance$(Q, t)$. □

**Theorem 2.61.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then* pathflux$(m \cdot P, N, N + m \cdot P, N + m \cdot P) = -1$.

*Proof.* The goal follows from periodicseedflux$(T, P, m, N)$. □

**Theorem 2.62.** leftphase$(3, 0, 3) = 1$ *and* leftphase$(3, 1, 2) = 0 \wedge$leftphaseflux$(3, 0, 2) = -1 \wedge$ leftgatephase$(3, 0, 2) = 1$.

*Proof.* The goal follows from seedphasecancellation. □

**Theorem 2.63.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $s$, if* rightmostleft$(Q, t, k)$*, then if $s < k$, then if* centercleanthrough$(Q, t, s)$*, then* rightmostleft$(Q, t + s, k - s)$.

*Proof.* The goal follows from rightmostleftprofile$(Q, t, k, s)$. □

**Theorem 2.64.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance$(Q, t, k)$*, then if* centercleanthrough$(Q, t, k)$*, then* firstpatherasure$(Q, t, k, k - 2)$ *and* erasureevent$(Q, t + (k - 1), 0) \wedge (\text{gatecode}(Q, t + (k-1), 0) = 2 \vee \text{gatecode}(Q, t + (k-1), 0) = 5) \wedge \text{center}(t + (k - 1)) = \text{false}$.

*Proof.* The goal follows from leadingfirsterasure$(Q, t, k)$. □

**Theorem 2.65.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b + r) =$ true*, then there exists a natural number $s$ such that there exists a natural number $k$ such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge k + s \leq r + 1 \wedge$* leadingadvance$(Q, b + s, k)$.

*Proof.* The goal follows from blackwindowforcesleadingsource$(Q, b, r)$. □

**Theorem 2.66.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k)$.

*Proof.* The goal follows from periodicleadingsourcewindow$(T, P, m, N)$. □

**Theorem 2.67.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k) \wedge$ firstpatherasure$(m \cdot P, t, k, k - 2) \wedge (\text{gatecode}(m \cdot P, t + (k-1), 0) = 2 \vee \text{gatecode}(m \cdot P, t + (k-1), 0) = 5)$.

*Proof.* The goal follows from periodicleadingerasurewindow$(T, P, m, N)$. □

**Theorem 2.68.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentleadingerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentleadingerasure$(T, P, m)$. □

**Theorem 2.69.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$,* leftphase$(Q+R, t, k) =$ leftphase$(Q, t, k)+$ leftphase$(R, t + Q, k)$.

*Proof.* The goal follows from leftphaselagadd$(Q, R, t, k)$. □

**Theorem 2.70.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$ and natural number $n$,* $\mathrm{pathflux}(Q + R, t, k, n) = \mathrm{pathflux}(Q, t, k, n) + \mathrm{pathflux}(R, t + Q, k, n)$.

*Proof.* The goal follows from $\mathrm{pathfluxlagadd}(Q, R, t, k, n)$. □

**Theorem 2.71.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leadingadvance}(Q, t, k)$*, then if* $\mathrm{centercleanthrough}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+k, 0) = 3 \leftrightarrow \mathrm{leftgatephase}(Q, t+(k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = -1$ *and* $\mathrm{leftcolour}(Q, t+k, 0) = 0 \leftrightarrow \mathrm{leftgatephase}(Q, t + (k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = 1$.

*Proof.* The goal follows from $\mathrm{leadingphasereset}(Q, t, k)$. □

**Theorem 2.72.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t + 1, 0) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, 0)$ *if and only if* $\mathrm{gatecode}(Q, t, 0) = 2$ *or* $\mathrm{gatecode}(Q, t, 0) = 5$.

*Proof.* The goal follows from $\mathrm{centererasureiffgate}(Q, t)$. □

**Theorem 2.73.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{gatecode}(Q, b + i, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + i, 0) = 5$*, then if* $\mathrm{gatecode}(Q, b + k, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + k, 0) = 5$*, then* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

*Proof.* The goal follows from $\mathrm{whiteblockgateconvex}(Q, b, n, i, j, k)$. □

**Theorem 2.74.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if* $\mathrm{centercleanthrough}(Q, b, n+1)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{erasureevent}(Q, b + i, 0)$*, then if* $\mathrm{erasureevent}(Q, b + k, 0)$*, then* $\mathrm{erasureevent}(Q, b + j, 0)$ *and* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

*Proof.* The goal follows from $\mathrm{whiteblockerasureconvex}(Q, b, n, i, j, k)$. □

**Theorem 2.75.** $\mathrm{leadingadvance}(4, 5, 2)$ *and* $\mathrm{centercleanthrough}(4, 5, 2) \wedge \mathrm{gatecode}(4, 6, 0) = 5$.

*Proof.* The goal follows from seedleadinggatefive. □

**Theorem 2.76.** $\mathrm{leadingadvance}(7, 7, 4)$ *and* $\mathrm{centercleanthrough}(7, 7, 4) \wedge \mathrm{gatecode}(7, 10, 0) = 2$.

*Proof.* The goal follows from seedleadinggatetwo. □

**Theorem 2.77.** leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ gatecode$(11, 7, 0) = 5 \wedge$ leftcolour$(11, 8, 0) = 3$.

*Proof.* The goal follows from seedleadingblackreset. □

**Theorem 2.78.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* recoverphase$(incomingBit, laterBit,$ transferphase$(incomingBi$ *incoming.*

*Proof.* The goal follows from transferrecover$(incomingBit, laterBit, incoming, gate)$. □

**Theorem 2.79.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* transferphase$(incomingBit, laterBit, incoming, gate) =$ $0$ *if and only if incoming* $=$ recoverphase$(incomingBit, laterBit, 0, gate)$.

*Proof.* The goal follows from transferkernel$(incomingBit, laterBit, incoming, gate)$. □

**Theorem 2.80.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t{+}1, k) =$ transferphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k)$, leftphase$(Q, t, k{+}1)$, leftgatephase$(Q, t, k)$).

*Proof.* The goal follows from phasetransfer$(Q, t, k)$. □

**Theorem 2.81.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k)$, leftphase$(Q, t{+}1, k)$, leftgatephase$(Q, t, k)$).

*Proof.* The goal follows from phaserecover$(Q, t, k)$. □

**Theorem 2.82.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$*, then not* leftgatephase$(Q, t, k) = 0$ *and* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k), 0,$ leftgatephase$(Q, t, k)$).

*Proof.* The goal follows from erasurerecovery$(Q, t, k)$. □

**Theorem 2.83.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then there exists a natural number $t$ such that $N \leq t$ and* erasureevent$(m \cdot P, t, 0) \wedge$ gatecode$(m \cdot P, t, 0) = g$.

*Proof.* The goal follows from periodicerasureorientation$(T, P, m, N, g)$. □

**Theorem 2.84.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance$(Q, t, k)$*, then if* centercleanthrough$(Q, t, k)$*, then there exists a natural number $s$ such that $s + 1 < k$ and* center$(t + s) =$ true*.*

*Proof.* The goal follows from leadingblackbeforearrival$(Q, t, k)$. □

**Theorem 2.85.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap$(b, r)$*, then if* leftadjacentdefect$(Q, b+ r) =$ true*, then there exists a natural number $k$ such that $k \leq r + 1$ and* leadingadvance$(Q, b, k)$*.*

*Proof.* The goal follows from blackgapforcesinitialleading$(Q, b, r)$. □

**Theorem 2.86.** *For every natural number $Q$ and natural number $t$, if* leadingadvance$(Q, t, 2)$*, then if* centercleanthrough$(Q, t, 2)$*, then* center$(t) =$ true*.*

*Proof.* The goal follows from leadingdepthtwoblack$(Q, t)$. □